

DAY 5 – OLAP CUBE & CUSTOMER 360 DATA WAREHOUSE

Technical Report

1. Introduction

This report summarizes the work implemented in the Day 5 OLAP Cube notebook. The notebook processes a Customer 360 dataset containing customer, order, product, support-ticket, and clickstream information. The workflow transforms the source data through cleaning and integration, designs a star-schema data warehouse, creates dimension and fact tables with surrogate keys, validates data quality, exports warehouse tables, prepares SQL table definitions, and finally simulates OLAP analysis using pandas pivot tables.

2. Objectives

- Load and inspect Customer 360 source datasets.
- Clean and standardize the source data and convert important fields to suitable data types.
- Create the derived order measure $\text{total_amount} = \text{quantity} \times \text{price}$.
- Integrate customer and order data into a `customer_orders` dataset.
- Design a star-schema data warehouse with dimension and fact tables.
- Generate surrogate keys and establish logical foreign-key relationships.
- Validate missing keys, duplicates, data types, quantities, prices, totals, and referential integrity.
- Export warehouse tables and generate SQL CREATE TABLE statements.
- Simulate an OLAP cube and demonstrate slice, dice, drill-down, and roll-up concepts.

3. Source Data and Initial Inspection

The notebook loads five CSV-based datasets: customers, orders, products, support tickets, and clickstream. The recorded source sizes are 3 customer rows, 4 order rows, 3 product rows, 2 support-ticket rows, and 4 clickstream rows.

Dataset	Rows	Columns
Customers	3	5
Orders	4	6 initially
Products	3	4
Support Tickets	2	4
Clickstream	4	4

The profiling results show zero missing values and zero duplicate rows in the source datasets. The customer table contains three unique customers, while the order data contains four unique order IDs. The source order dates were initially stored as object values and were later converted to datetime.

4. Data Preprocessing and Integration

- Customer column names were standardized using lowercase and whitespace stripping.

- Duplicate customer rows were removed.
- Customer names were stripped of surrounding whitespace.
- order_date was converted to datetime using `pd.to_datetime(..., errors='coerce')`.
- quantity was converted to numeric.
- total_amount was derived from quantity × price.
- customers and orders were merged on customer_id using a left join.

The resulting customer_orders dataset contains 4 rows and 11 columns. The validation output reports zero missing values and zero duplicate rows after integration. The integrated table contains customer attributes together with order, product, date, quantity, price, and total_amount information.

5. Data Warehouse Design – Star Schema

The notebook designs a star schema in which fact tables represent business events and measures, while dimension tables provide descriptive context.

Table	Type	Purpose
dim_customer	Dimension	Customer demographic and identifying information
dim_product	Dimension	Product name and category information
dim_date	Dimension	Calendar attributes for time-based analysis
dim_support_status	Dimension	Standardized support-ticket statuses
dim_page	Dimension	Pages visited in the clickstream
fact_orders	Fact	Order-level measures such as quantity, price, and total_amount
fact_support_tickets	Fact	Support-ticket events and customer/status keys
fact_clickstream	Fact	Page-visit events linked to customer, date, and page

6. Surrogate Keys and Fact Table Construction

Surrogate keys were generated as sequential integer keys for the dimension tables while retaining the original business keys for traceability. The dimensions produced include 3 customers, 3 products, 2 support statuses, 3 pages, and a date dimension. The notebook also creates a continuous date dimension covering the order period from 2023-01-01 through 2023-01-10.

- fact_orders: 4 rows × 7 columns.
- fact_support_tickets: 2 rows × 4 columns.
- fact_clickstream: 4 rows × 5 columns.
- fact_orders links customer_key, product_key, and date_key to their corresponding dimensions.
- fact_support_tickets links customer_key and support_status_key.
- fact_clickstream links customer_key, date_key, and page_key.

7. Validation and Data Quality

The validation stage checks surrogate-key completeness, dimension-key uniqueness, fact-table duplicates, and critical data types. The recorded validation results are clean.

Validation Check	Result
Missing surrogate keys in fact tables	0
Duplicate surrogate keys in dimensions	0
Duplicate fact rows	0
Duplicate order IDs	0
Negative quantities	0
Negative prices	0
Negative total amounts	0
Invalid customer foreign keys	0
Invalid product foreign keys	0
Invalid date foreign keys	0

An additional calculation confirms that every recorded total_amount equals quantity multiplied by price; all four comparison differences are 0.0.

8. OLAP Cube Simulation

The notebook simulates an OLAP cube with pandas pivot_table. The main measure is total_amount, and the analysis combines customer, product category, and time dimensions.

Analysis	Dimensions	Measure	Recorded Result
Sales by customer and category	Customer + Gender + Category	Sum of total_amount	Alice: Books 40, Electronics 50; Bob: Electronics 50; Charlie: Home Goods 150
Sales by year and month	Year + Month	Sum of total_amount	2023 January = 290
Average order value by age group	Age group	Mean total_amount	<25 = 50; 25–34 = 45; 35–44 = 150

9. OLAP Operations Demonstrated

- Slice: restrict the cube to a particular dimension value, such as a selected product category.
- Dice: select a subset using values across multiple dimensions, such as electronics sales.
- Drill Down: move from a higher-level time aggregation toward a more detailed level such as day.
- Roll Up: aggregate detailed data toward a higher level such as year.

10. Export and SQL Preparation

The notebook exports warehouse tables to CSV and generates SQL CREATE TABLE statements. The SQL generator maps pandas integer, floating-point, string, datetime, and Boolean types to INTEGER, REAL, VARCHAR(255), TIMESTAMP, and BOOLEAN respectively. Primary and foreign-key constraints are included in the generated definitions.

11. Findings and Interpretation

- The four orders generate total sales of 290.0 in the recorded OLAP analysis.
- Charlie contributes the largest individual order value, with 150.0 from Product C in Home Goods.
- Alice has two orders totaling 90.0: 50.0 in Electronics and 40.0 in Books.
- Product A contributes 100.0 across Alice and Bob.

- The 35–44 age group has the highest recorded average order value at 150.0, based on the small sample.
- The dataset is very small and represents demonstration data, so these patterns should not be treated as general business trends.

12. Limitations and Notes

The notebook is primarily a demonstration implementation. The support-ticket source does not contain a ticket creation date, so support events cannot be linked to the date dimension in the current design. The notebook also uses a placeholder `is_holiday` value in one version of the date dimension. In addition, different stages show both a distinct-date version and a continuous-date version of `dim_date`; the continuous version contains 10 daily records for the 2023-01-01 to 2023-01-10 range.

The final OLAP section operates successfully on the order-oriented warehouse tables and demonstrates multi-dimensional aggregation through pandas pivot tables. For a production warehouse, the schema would normally be implemented in a database or analytical platform with enforced constraints and a fuller calendar.

13. Conclusion

The Day 5 implementation successfully demonstrates an end-to-end ETL and OLAP workflow for a Customer 360 dataset. Raw customer, order, product, support, and clickstream data are profiled and cleaned, integrated into a warehouse-oriented structure, organized into dimension and fact tables, linked with surrogate keys, validated for data quality, exported, and prepared for relational database loading. The OLAP simulation then provides multi-dimensional sales analysis by customer, product category, time, and customer age group. Overall, the notebook provides a practical foundation for analytical reporting and demonstrates how a star schema can support flexible business analysis.

14. Source

This report was prepared directly from the uploaded notebook: `Day_5_OLAP_Cube.ipynb`. The report preserves the notebook's terminology and recorded outputs rather than introducing external data.